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How important is the correlation between returns and volatility in a stochastic volatility model? Empirical evidence from pricing and hedging in the S&P 500 index options market

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Abstract

This paper examines the importance of allowing for correlation between returns and volatility in a continuous time stochastic volatility option pricing model. Specifically it tests the closed-form stochastic volatility model of Heston (Review of Financial Studies 6, 1993, 327–343) that allows for non-zero correlation, in terms of pricing and hedging options on the S&P 500 index. It is found that non-zero correlation in the stochastic volatility model leads to significant improvements in mispricing of out-of-the-money options and overall pricing performance compared to if correlation is constrained to be zero. In terms of hedging, non-zero correlation results in significantly lower hedging errors for out-of-the-money options. © 1998 Elsevier Science B.V. All rights reserved.

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1. Introduction

It is well known that volatility of the underlying asset price (as measured by its standard deviation or variance) is one of the primary determinants of option prices and hedge ratios. An option pricing model that does not properly capture the evolution of volatility process can give rise to option prices that do not agree well with prices observed in the market and could also deteriorate an investor's ability to hedge risk. Rubinstein (1985, 1994) documents various option pricing anomalies in the Black and Scholes (1973), henceforth BS, framework in the equity and index options market, the most notable of which is the volatility smile evident in option prices.² Several others have documented similar biases and anomalies in other financial markets, a comprehensive discussion of which can be found in Bates (1995).

In trying to address these biases, researchers have tried to relax one of the crucial assumptions of the BS framework, namely that volatility evolves deterministically. One strand of the literature has assumed that volatility evolves as a separate process in itself, in that the innovations governing the evolution of the volatility process are different from the innovations governing the asset returns process, though the two may be correlated. These models are often referred to as stochastic volatility (henceforth SV) models of option pricing and have been mostly developed in continuous time, in that trading is assumed to take place continuously through time. On the other hand, the GARCH models of volatility, developed in discrete time assume that the innovations driving the asset returns and volatility are the same. GARCH option pricing models have been studied by Amin and Ng (1993), Duan (1995) and Engle and Mustafa (1992) and Engle et al. (1994, 1997). It is interesting to note that Nelson (1990) and Foster and Nelson (1994) have shown that the discrete time GARCH models converge to continuous time stochastic volatility models which implies that the two models are related.

The issue of SV and its effects on option prices have been studied extensively. Some of the earlier work in this area include Johnson and Shanno (1987), Hull and White (1987), Scott (1987), Wiggins (1987), Chesney and Scott (1989) and Melino and Turnbull (1990). The models developed by these authors require either the use of Monte-Carlo simulation or the numerical solution of a two dimensional parabolic partial differential equation and are computationally demanding. Many of these models make the questionable assumption that asset returns and volatility are uncorrelated and some assume that the volatility risk is not priced. Stein and Stein (1991) provide a more

² The fact that as the strike price of an option increases, its implied volatility (from the Black–Scholes model) decreases is often referred to as the volatility smile, although the curve looks more like a smirk.

efficient solution for option prices that can be got via bivariate numerical integration; however their model is also developed under the unrealistic assumption of zero correlation between spot asset returns and the volatility process.

While the assumption that asset returns and volatility are uncorrelated in a SV framework may be tenable in some markets, it is clearly unrealistic in the market for index options as evident in Rubinstein (1994), who finds that the local volatility of the index (implied using option prices) is negatively correlated with the level of the index. The smile (or more appropriately the smirk) feature in implied volatilities stems from negative skewness in the risk-neutral distribution of the asset returns. In a SV model, without jumps in the underlying asset's returns, this negative skewness can only be achieved through a negative correlation between returns and volatility.³ Estimating the correlation from the time series of asset returns (Chesney and Scott, 1989; Melino and Turnbull, 1990) identifies the correlation between asset returns and historical volatility which could be very different from the correlation between asset returns and implied volatility embedded in market option prices.

Heston (1993) started a new approach to option pricing under stochastic volatility. The methodology used therein yielded closed-form solution for European option prices and at the same time permitted a non-zero risk premium for volatility as well as arbitrary correlation between asset returns and volatility.⁴ Furthermore, Heston's approach also permits closed form solutions for the option's delta and vega. Consequently, this framework enables one to test in an efficient manner, both the pricing and hedging effectiveness of a SV model without being encumbered by the additional complications of other SV models and the possible clouding of inference therein. In particular, one could use the information contained in a long time series (with a representative broad cross-section) of option prices to extract the model parameters in an in-sample context and thereafter compute out-of-sample option prices and hedge ratios for testing the efficacy of an option pricing model. Given existing technology, this is not readily feasible under the earlier genre of SV models, many of which are predicated on the absence of any correlation between returns and volatility.

This paper examines the marginal contribution of allowing for non-zero correlation in a SV model by utilizing the information implicit in option prices and at the same time, provides an empirical test of the pricing and hedging effectiveness of the SV model developed by Heston (1993) in the S&P 500 index options

³ In a SV model, one can always find a risk-neutral distribution, except that unlike in a deterministic volatility model, the distribution is not unique and depends on the utility function of a representative investor.

⁴ Bates (1996a, b) and Scott (1997) extended Heston's framework to allow for jump diffusion in asset returns and stochastic interest rates.

market. Specifically, it tests two versions of the model, one in which correlation is constrained to be zero and another in which correlation is a free parameter. The time invariant parameters of the model are estimated jointly with the time varying volatility from a time series (and cross-section) of option prices without having to re-estimate the model daily as in Bakshi et al. (1997) or constructing a time series of variance from the Black–Scholes implied volatilities as in Knoch (1992). Implementation of the hedging test takes into account the round trip transaction costs in the options market and S&P 500 futures are used instead of the level of the index, thus conforming to reality as much as possible. Bates (1996a, b) provides an empirical examination of Heston's SV framework in the currency options and S&P 500 futures options market as a subset of a jump diffusion-SV model. However, the empirical focus of these two papers is to examine the consistency of the return and volatility distributions inherent in market option prices with the observed time series properties of asset returns and implied volatilities. He does not examine the out-of-sample pricing and hedging performance of the model, the focus of this paper.

It is found that accounting for correlation between returns and volatility in the SV model substantially improves the mispricing of out-of-the-money options, compared to both the zero correlation version of the SV model and the widely used version of the BS model in which an implied volatility is estimated daily. Surprisingly, the version of the SV model with zero correlation under-performs in terms of pricing (out-of-sample) deep out-of-the-money calls and puts. This result runs contrary to the typical belief that a SV model with (even without correlation) is always more desirable than the BS model and underscores the importance of allowing for correlation in a SV model. In terms of hedging, allowing for correlation in the SV model results in lower (statistically significant) root mean squared errors for out-of-the-money options. Since out-of-the-money puts are heavily traded in the index options market, this result has some practical relevance.

Section 2 describes the model, Section 3 describes the methodology and data, Section 4 describes the results and Section 5 concludes.

2. Model

Let the spot asset price, S_t and the spot variance, v_t follow the following two processes:

$$dS(t) = \mu S \, dt + \sqrt{v(t)} S \, dz_1(t), \quad (1)$$

$$dv(t) = \kappa[(\theta - v(t))] \, dt + \sigma \sqrt{v(t)} \, dz_2(t), \quad (2)$$

where z_1 and z_2 are two Wiener processes with correlation ρ . The asset price evolves as a geometric Brownian motion and the variance of the instantaneous

returns v_t evolves as a square root process with long-term mean θ .⁵ The speed with which v_t reverts towards θ is measured by κ . It can be shown (see the appendix) that the half life of the volatility process is $\ln(2)/\kappa$.

Given the dynamics in Eqs. (1) and (2), Heston (1993) shows that the price at time t of a European call option that matures at T ($\tau \equiv T - t$) and has an exercise price of K is given by

$$C(S, v, t) = SP_1 - Ke^{-r_f\tau}P_2, \quad (3)$$

where

$$P_j(x, v, \tau; \ln(K)) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left(\frac{e^{-i\phi \ln(K)} f_j(x, v, \tau; \phi)}{i\phi} \right) d\phi, \quad (4)$$

where $\operatorname{Re}(\cdot)$ denotes the real part of a complex variable, $x \equiv \ln(S)$, $f_j(x, v, \tau; \phi) = \exp(C_j(\tau; \phi) + D_j(\tau; \phi)v + i\phi x)$, $C_j(\cdot)$ and $D_j(\cdot)$ are functions of $\kappa\theta$, $\kappa + \lambda$, ρ , σ , r_f and τ and λ is a constant such that λv_t measures the volatility risk premium.

Since the option price is homogeneous of degree 1 in S_t and K , it is easy to see that $C_t = S_t(\partial C_t / \partial S_t) + K(\partial C_t / \partial K)$. Therefore it follows from Eq. (3) that the delta of the option is $P_1(\cdot)$. Differentiating with respect to v , the vega of the option is given by

$$\frac{\partial C}{\partial v} = \frac{S}{\pi} v_1 - \frac{Ke^{-r_f\tau}}{\pi} v_2, \quad (5)$$

where

$$v_j = \int_0^\infty \operatorname{Re} \left(\frac{D_j(\cdot) \exp(-i\phi \ln(K) + C_j(\cdot) + D_j(\cdot)v + i\phi x)}{i\phi} \right) d\phi. \quad (6)$$

The option price and hedge ratios are functions not only of S_t , v_t , r , K , τ (that appear in the BS model, but also of $\kappa + \lambda$, $\kappa\theta$, ρ and σ that are unobservable and have to be estimated. $\kappa + \lambda$ and $(\kappa\theta)/(\kappa + \lambda)$ are the mean reversion parameter and the long run mean of the risk neutralized variance process respectively.

⁵ The variance, v_t follows a square root diffusion process and is therefore precluded from being negative (see Cox et al., 1985).

3. Data and methodology

3.1. Data

Intra day data on S&P 500 index options (ticker symbol SPX) that are traded on the Chicago Board Options Exchange (CBOE) are used to test the model. The market for S&P 500 index options is the second most active index options market in the United States and, in terms of open interest in options, it is the largest. The options are European and settled for a cash amount equal to 100 times the difference between the index level and strike price. Unlike options on the S&P 100 index, there are no wild card features that can complicate valuation. Also there is a very active market for the S&P 500 futures, making the replication of the index much easier. Thus, according to Rubinstein (1994) it is one of the best markets for testing a European option pricing model.

The following rules are applied to filter data needed for the empirical test.

1. For each day, a bid–ask quote has to occur between 2:00 and 2:20 p.m. The tight time window is chosen to minimize problems stemming from intraday variation in volatility.⁶
2. An option of a particular moneyness and maturity is represented only once in the sample. In other words, although the same option may be quoted again during the time window, only the first record of that option is included in our sample.
3. A transaction has to satisfy the no-arbitrage relationships as outlined in Propositions 1 and 2 of Cox and Rubinstein (1985, pp. 129–133).⁷
4. The mid point of the bid–ask spread has to be at least 50 cents to reduce problems arising from minimum bid–ask spreads (that can constitute a big percentage of price) in low priced options.
5. All transactions that were executed electronically are excluded.⁸

The sample is created such that an option appears in the chosen time window on at least two consecutive days. This is because the estimation procedure (discussed in Section 3.2) takes into account the autocorrelation in the option pricing errors for which an option has to be present on at least two consecutive days.

Stale index levels of the S&P 500 (stemming from infrequent trading of the constituent securities) could potentially be a problem. One must note that the S&P 500 is a value weighted index and the bigger stocks that trade more

⁶ We also used two other windows, namely between 9:00 and 9:30 a.m. as well as between 2:30 and 3:00 p.m. However, the primary results of the paper are invariant with respect to the time window used.

⁷ The propositions are adapted from Merton (1973).

⁸ Only a small fraction of the trades is electronically executed in the time period we looked at.

frequently constitute the bulk of the index level. Since intraday data and not the end-of-the-day option prices are used, the staleness problem may not be severe enough to undermine the estimation procedure. In theory, one could possibly overcome this problem by using implied index levels from the put–call parity equation. However, this is conditional on put call parity holding as an equality and in the presence of transactions costs (bid–ask spreads that are non-negligible), the equality becomes an inequality. Thus the implied index levels from the put–call parity equation may not equal the true index level. Also, even if one assumes away transactions costs, it is very difficult to create a sample of sufficient size by creating matched pairs of puts and calls because the level of the S&P 500 index changes quite frequently through the day. As a result, the index levels as reported in the data set are used in parameter estimation. However, in the hedging test, S&P 500 futures data is used instead of the spot level of the index. The daily dividend yield on the S&P 500 index are taken from Data Stream. For the risk free rate, a time series of bond yields from the daily treasury yield curves prepared by Salomon Brothers and interpolated to match the yield to the expiration date of the option is used. The filtered data set on which the estimation of the parameters of the SV model is carried out consists of 9548 records, from 21 January 1991 to 10 April 1992, a total of 310 trading days. Of these 3568 are calls and 5980 are puts.

3.2. Methodology

Central to the empirical test is the estimation of the unobservable parameters, $\kappa + \lambda$, $\kappa\theta$, ρ , σ and v_t for the stochastic volatility model that appear in the pricing and hedging formulae.⁹ Since the spot variance, v_t , is unobservable, the estimation of these parameters from a time series of v_t is difficult. Although one has high frequency intra day data, the construction of a daily v_t series from the intra day data is subject to question due to the bid–ask bounce present in the intraday data. The moments of the log returns can be calculated using the formula of Heston (1993) for the characteristic function of the log spot price. However, Kearns (1992) has shown that using generalized method of moments (Hansen, 1982) to estimate parameters from the asset returns in a SV model data is sensitive to the choice of moments. Foster and Nelson (1994) describe a procedure to estimate the parameters (i.e., κ , θ , ρ , σ) of the unobservable volatility process from the returns data. Unfortunately, their procedure to estimate the parameters from the daily S&P 500 returns did not work on this data set in that numerical problems were encountered. Further, this procedure

⁹ Note that one does not have to separately estimate κ and θ because the pricing and hedging formulae are in terms of $\kappa + \lambda$ and $\kappa\theta$.

is not able to identify λ that has to be estimated from the options data conditional on the estimates of other parameters.

Since closed-form solutions are available for the option price, a natural candidate for the estimation of the parameters which enter the pricing and hedging formulae is a non-linear least squares procedure involving minimization of the sum of squared errors between the model and market prices. The primary advantage of using market option prices to estimate the parameters is that it allows one to use the information contained in the market option prices, information that essentially is forward looking.

Let $O_{i,t}$ denote the market price of option i on day t and $f_{i,t}(\Psi, v_t)$ denote the model price of the option i at time t as a function of the set of parameters Ψ, v_t , where $\Psi \equiv (\kappa + \lambda, \kappa\theta, \rho, \sigma)$ for the SV model. It is assumed that the spot variance, v_t does not vary between 2:00 and 2:20 p.m., the time period used to create the sample of options every day. v_t is a parameter to be estimated and allowing v_t to change every time t changes will increase the number of parameters by thousands given the size of the data set

Let the option pricing error be

$$e_{i,t}(\Psi, v_t) = O_{i,t} - f_{i,t}(\Psi, v_t). \quad (7)$$

If e_{it} are i.i.d., the criterion function is

$$\min_{\Psi, v_t} \sum_{t=1}^T \sum_{i=1}^{N_t} e_{i,t}^2(\Psi, v_t), \quad (8)$$

where T is the number of trading days in the sample and N_t is the number of options on a particular day t in the sample. However, the above criterion function cannot be used if the errors are autocorrelated. Autocorrelation in residuals can arise from a misspecification of the model, such as the omission of a relevant state variable or from bid–ask bounce. Since all models are simplifications of the real world, autocorrelation in the residuals is an important issue that ought to be addressed in our testing. A preliminary estimation of the model using Eq. (8) and a Box–Jenkins analysis indicates that an AR (1) process parsimoniously captures the evolution of the pricing errors and the autocorrelation coefficient depends on the moneyness of an option. Therefore, we assume a separate autocorrelation coefficient for near-the-money, in-the-money, deep-in-the-money, out-of-the-money and deep-out-of-the-money options. In particular, it is assumed that the pricing errors for each of these five categories of options evolve as follows:

$$e_{i,t} = \rho_i^* e_{i,t-1} + v_{i,t}. \quad (9)$$

$e_{i,t}$ denotes the pricing error for an option belonging to moneyness class i at time t and the variance–covariance matrix of v is $E(v_{i,t} v_{j,t}) = \phi_{i,j}$. Given this, we have a non-linear SURS (seemingly unrelated regressions) framework with

five cross-sections every day and it can be shown that the appropriate sum of squared errors is (Judge et al., 1985, pp. 483–490)

$$S(\Psi, \rho^*, v_t) = \sum_{i=1}^5 \epsilon_i^2 + \sum_{t=2}^T \sum_{i=1}^5 [O_{i,t} - f_{i,t}(\Psi, v_t) - \rho_i^*(O_{i,t-1} - f_{i,t-1}(\Psi, v_{t-1}))]^2. \quad (10)$$

ϵ_i represents the transformed errors for moneyness class i for the first day and a particular ϵ_i depends on the other i s as well, the relationship being determined by the variance–covariance matrix of v as well as each ρ_i^* (Judge et al., 1985, pp. 483–490). Note that i in Eq. (10) varies from 1 to 5 (and not from 1 to N_t as in Eq. (8)) because individual options have been pooled to belong to five separate moneyness classes. Eq. (10) is the total SSE for the entire sample and is the sum of the SSE of the individual days. The first expression in the RHS of Eq. (10) represents the SSE for day 1 and the second expression represents the sum (over days) of the individual SSE of each day, starting from day 2. In such a framework estimating Ψ simultaneously with each ρ_i^* is very computationally intensive as it amounts to doing full information maximum likelihood. Therefore, as suggested in Judge et al. (1985), we adopt a two-stage procedure. In the first stage we estimate (Ψ, v_t) by minimizing the sum of squared errors for each moneyness class individually under the i.i.d. assumption. Thereafter, ρ^* for each moneyness class and the variance–covariance matrix of v are estimated from the residuals of the first stage. In the second stage, we estimate Ψ and v_t simultaneously from Eq. (10), conditional on the estimated ρ^* and the variance–covariance matrix of v .

Since the daily spot variance, v_t is a parameter to be estimated, the number of parameters to be estimated is proportional to the number of days in the sample. To get around the problem of estimating too many parameters, one could possibly estimate v_t and Ψ (treating ρ^* to be zero) every day from the daily minimization of the sum of squared errors between model and market option prices as done in Bakshi et al. (1997). In this way one has a different Ψ and v_t every day. However, it must be noted that according to the model, Ψ is constant and only v_t varies. Our approach treats (Ψ, ρ^*) to be constant through the sample (i.e., the same Ψ is used $\forall t$) and v_t is estimated from the cross-section of option prices every day.¹⁰ $\kappa + \lambda$, $\kappa\theta$ and v_t are constrained to be non-negative by an exponential transformation.¹¹ ρ is constrained between -1 and 1 by a sine/cosine transformation. The SV model price for the option is evaluated by Gaussian quadrature (with hermite polynomials) and the number of quadrature

¹⁰ The basic NLS routine used is Marquardt's method as given in Press et al. (1992).

¹¹ $\kappa + \lambda$ and $\kappa\theta$ have to be non-negative for the long-run mean of the risk neutralized variance process to be non negative.

points used is 80.¹² Derivatives of the model option price with respect to Ψ are evaluated numerically. The results of the estimation for call and put options are discussed in Section 4.

3.3. Hedging

A complete test of an option pricing model should take into account not only how well the model prices match observed option prices, but also, and equally importantly, the ability of the model to hedge against changes in the underlying state variables. The hedging issue is of interest not only to academics, but also to actual traders in the option market who often have to hedge their positions in the underlying asset or options market. In the context of the SV model, it is important to see how the incorporation of correlation between index returns and volatility change returns of a hedge portfolio. Since the underlying asset is an index of 500 stocks, in order to perform a realistic hedging test, we choose the S&P 500 futures and not the level of the index (at a given time in the day) from the options data set. It is to be noted that we had the matching data set for both options and futures only for a limited period of time. In fact, when we estimated the parameters, we used the level of the index (and not the implied level of the index from the futures data) from the options data set simply because we did not have the futures data set for the time period that was used in the in-sample estimation.

In the BS model of option pricing, if one creates a portfolio consisting of a long position in a certain number of units of the underlying asset (as given by the option's delta) and short one unit of the call option, then the portfolio has no risk over the next instant. In the BS model, there is only one state variable (the asset price) and one can create a hedge portfolio by trading only in the underlying asset and one option. However, if there are two state variables as in the SV model, then one more option is needed to hedge the additional volatility risk and three assets are needed to form an instantaneously risk-free portfolio. The portfolio that we form consists of w_1 units in the S&P 500 futures, w_2 units in one option and a single unit in a different option. w_1 and w_2 are easily found by solving two equations and one can refer to Scott (1987) for an illustration; w_1 and w_2 are functions of deltas and the vegas of the two options and are therefore available in closed-form.

In order to perform the out-of-sample hedging analysis, v_t is estimated each day by minimizing the sum of squared errors between model and market prices,

¹² For most option records, we need a maximum of 20 quadrature points to arrive at a reliable option price. However, experimentation with artificial data indicates that a greater number of quadrature points is needed for out-of-the-money options. The quadrature routine that we use is "gauher" in Press et al. (1992).

conditional on the time invariant parameters that have been estimated from the past data. The way a time series of returns on the hedge portfolio is created is as follows: On day t (given Ψ and v_{t-1} for the SV model and the previous day's implied volatility for the BS model), two options belonging to a particular moneyness/maturity class are selected from the first bid–ask quotes in those options between 2:00 and 2:20 p.m. by buying/selling at the ask/bid. Immediately after that, one takes a long position in S&P 500 futures by selecting the next available futures transaction (from the futures data set). On day $t + 1$, the value of the portfolio is calculated from the first bid–ask quotes in those options between 2:00 and 2:20 p.m. and the next available futures transaction. The values of the option positions on day $t + 1$ are calculated by liquidating the previous long/short position at the bid/ask, thus recognizing the round trip transactions costs in the options market. The normalized change in the hedge portfolio is defined to be the ratio of the change in the portfolio value to the value of the portfolio on day t .¹³

If a model hedges well, one would expect the normalized changes in the value of the corresponding hedge portfolio to very small (close to the overnight risk free rate). As a result, one would expect the time series of these changes to have very low variance and a mean close to zero. Therefore we compare the hedging performance of the two versions of the SV model (zero correlation and non-zero correlation) by comparing how the normalized changes of the respective hedge portfolios deviate from zero. Specifically, we compute the mean of the squared changes, that we refer to as the mean squared error (MSE), in the values of the hedge portfolios and the two models are compared in terms of the MSE of their respective hedge portfolios. Note that we could also compare the two models in terms of the variance of the normalized changes. However, given that the mean of these changes is very low, MSE and variance essentially convey the same message. The discrete rebalancing interval introduces an additional error term that has the effect of making the returns of the hedge portfolio heteroskedastic (Boyle and Emanuel, 1980), and possible model misspecification could introduce autocorrelation in the portfolio returns. Therefore, a GMM statistic is used to compare the normalized changes of the hedge portfolios of the two versions of the SV model and is discussed below.

Let $r_{sv_{\rho=0,t}}$ and $r_{sv_{\rho \neq 0,t}}$ denote the normalized changes of the hedge portfolios corresponding to the two versions of the SV models respectively. We want to test whether

$$E(r_{\rho=0,t})^2 - E(r_{\rho \neq 0,t})^2 = 0 \quad \text{or} \quad E[(r_{\rho=0,t})^2 - (r_{\rho \neq 0,t})^2] = 0. \quad (11)$$

¹³ With futures as part of the portfolio, returns of the hedge portfolios are not well defined. Hence we use the term, normalized change.

Let $Z_t \equiv (r_{\rho=0,t})^2 - (r_{\rho \neq 0,t})^2$. Then we want to test whether $E(Z_t) = 0$. A GMM statistic (in the Newey–West framework) for testing this is given by

$$V = \sqrt{T} \frac{\sum_{t=1}^T (Z_t - \mu_z)}{\sqrt{\sum_{t=1}^T (Z_t - \mu_z)^2 + 2 \sum_{j=1}^m \sum_{t=j+1}^T (Z_t - \mu_z)(Z_{t-j} - \mu_z)}}, \quad (12)$$

where μ_z represents the sample mean of Z_t and V is distributed asymptotically as a standard normal. The choice of m is somewhat ad hoc. However, different values of m are tested with and the results discussed in the following section remain essentially unchanged.

4. Results

4.1. Parameter estimates

As described before, we test two versions of the SV model, one in which the correlation between returns and volatility is a free parameter and another in which the correlation is constrained to be zero. The estimates of the various parameters of the two versions are reported in Tables 1 and 2. Note from these two tables that the autocorrelation coefficients of the option pricing errors are significantly positive for all moneyness classes.

The in-sample sum of squared errors is considerably higher under the constrained version of the SV model (i.e., in which ρ is set equal to zero) than the unconstrained version, which is to be expected. For the unconstrained model, ρ is -0.79 indicating that implied volatility and index returns, as embedded in the options data are highly negatively correlated. Volatility of variance, as

Table 1

Non-linear least squares parameter estimates of the stochastic volatility model in which the correlation between returns and volatility is unconstrained using daily option prices between 2:00 and 2:20 p.m. from 21 January 1991 to 10 April 1992

SSE	ρ_{DI}^*	ρ_I^*	ρ_N^*	ρ_O^*	ρ_{DO}^*	$\kappa\theta$	$\kappa + \lambda$	σ	ρ
4235.35	0.45	0.22	0.18	0.19	0.31	0.092	3.29	0.26	-0.79
	(0.25)	(0.27)	(0.11)	(0.11)	(0.14)	(0.0011)	(0.187)	(0.009)	(0.016)

Mid-point of bid and ask is used as the option price Total number of observations = 9548. Number of calls = 3568 and number of puts = 5980. ρ_{DI}^* , ρ_I^* , ρ_N^* , ρ_{DO}^* and ρ_O^* are the autocorrelation coefficients of the pricing errors of deep-in-the-money (612 options), in-the-money (1880 options), near-the-money (2168 options), deep-out-of-the-money (2060 options) and out-of-the-money (2828 options) options respectively. An option is near-the-money if $|(S - K)/S| \leq 0.01$. A call is deep out-of-the-money if $(K - S)/S > 0.04$, out of the-money if $0.01 < (K - S)/S \leq 0.04$. A call is deep in-the-money if $(S - K)/S > 0.04$ and in-the-money if $0.01 < (S - K)/S \leq 0.04$. Similarly the moneyness for puts is defined. Standard errors appear in parentheses and SSE is the sum of squared pricing errors.

Table 2

Non-linear least squares parameter estimates of the stochastic volatility model in which the correlation between returns and volatility is constrained to be zero

SSE	ρ_{DI}^*	ρ_I^*	ρ_N^*	ρ_O^*	ρ_{DO}^*	$\kappa\theta$	$\kappa + \lambda$	σ
5315.78	0.57 (0.29)	0.42 (0.25)	0.27 (0.14)	0.26 (0.12)	0.76 (0.42)	0.23 (0.0028)	8.64 (0.44)	1.28 (0.04)

measured by σ is considerably higher if we do not allow for correlation. A rough estimate of σ , given a discrete time series of v_t is ¹⁴

$$\sqrt{\frac{(\sum_{t=1}^{t=T} (v_t - v_{t-1}))^2}{\sum_{t=1}^{t=T} v_t}}.$$

Therefore one can perform a cross-check (somewhat crude but nevertheless useful) by comparing σ from the estimated time series of v_t with the σ that has been estimated directly from the options data. σ as implied by the options data is around 0.26 under the constrained model and 1.28 under the unconstrained model. While, neither the constrained, nor the unconstrained model's σ is consistent with the estimated v_t series in Fig. 1, the estimated σ when correlation is constrained to be zero is indeed highly implausible.¹⁵ This type of inconsistency between the volatility of variance implied directly as a parameter from the options data and the volatility computed from the time series of implied spot variance, v_t , has also been noted by Bates (1996a) for currency options under the square root variance model.

4.2. Out-of-sample pricing

The results of out-of-sample pricing from 13 April 1992 to 28 October 1992 (126 days) are shown in Table 3 for two versions of the SV model and the BS model. For the SV models, conditional on the estimates of the time invariant parameters obtained from the previous period, a daily variance, v_t is computed by minimizing the SSE between model and market prices (mid-points of bid–ask quotes between 2:00 and 2:20 p.m.), while for the BS model a single implied volatility is computed along similar lines. The implied volatility on day t is then used to price the options on day $t + 1$ for all options.

Table 3 reports the absolute out-of-sample percentage mispricing for the entire sample as well as within each option category classified by moneyness and

¹⁴ The rough estimate of σ is computed from the fact that $(dv_t)^2 = \sigma^2 v_t dt$ and approximating the integration in both sides by summation.

¹⁵ For sake of visual clarity, the estimated $\sqrt{v_t}$ series in Fig. 1 are from the model with non-zero correlation. However, the estimated $\sqrt{v_t}$ series from the model with zero correlation is very close to the series shown.

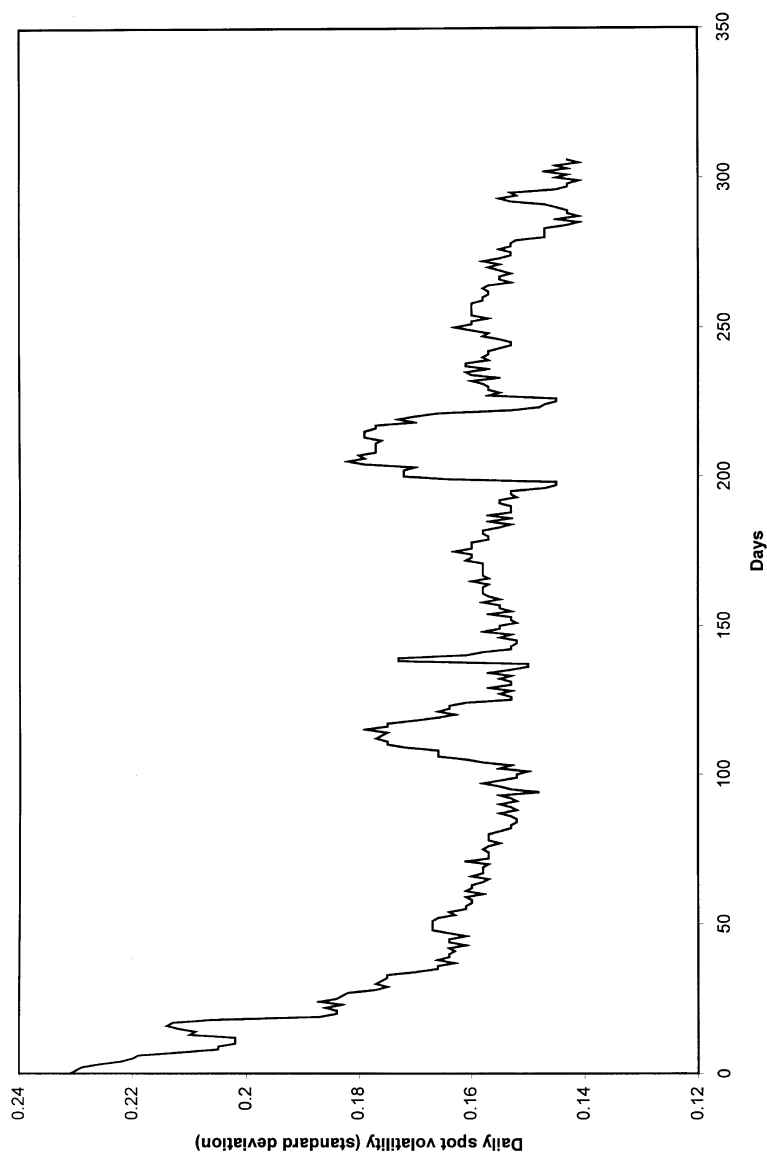


Fig. 1. Shows the estimates of the daily spot standard deviation inferred from options under the stochastic volatility model with non-zero correlation from 21 January 1991 to 10 April 1992.

Table 3

This table shows the out-of-sample average mispricing of different categories of options by money-ness and maturity in the time period, 13 April 1992–28 October 1992

	Moneyness	BS	$SV_{\rho=0}$	$SV_{\rho \neq 0}$
Maturity of calls (0–3 months)	< -0.04	0.025	0.023	0.011
	(-0.04, -0.01)	0.045	0.036	0.03
	(-0.01, 0.01)	0.191	0.136	0.089
	(0.01, 0.04)	0.48	0.36	0.24
	> 0.04	0.84	0.99	0.35
Maturity of calls (3 months and above)	< -0.04	0.044	0.025	0.021
	(-0.04, -0.01)	0.031	0.041	0.043
	(-0.01, 0.01)	0.061	0.042	0.039
	(0.01, 0.04)	0.136	0.051	0.048
	> 0.04	0.496	0.604	0.181
Average % mispricing		9.88%	7.16%	5.29%
Maturity of puts (0–3 months)	< -0.04	0.36	0.61	0.18
	(-0.04, -0.01)	0.11	0.17	0.08
	(-0.01, 0.01)	0.11	0.11	0.05
	(0.01, 0.04)	0.081	0.046	0.042
	> 0.04	0.035	0.03	0.01
Maturity of puts (3 months and above)	< -0.04	0.315	0.358	0.148
	(-0.04, -0.01)	0.121	0.173	0.071
	(-0.01, 0.01)	0.045	0.082	0.023
	(0.01, 0.04)	0.042	0.039	0.026
	> 0.04	0.049	0.079	0.021
Average % mispricing		11.68%	10.09%	5.3%

Mispricing for a single option is defined to be $|\text{Model Price} - \text{Market Price}| / \text{Market Price}$ and the market price is measured to be the mid point of bid–ask. The average mispricing for one particular category is the sum of the individual mispricings in that category divided by the total number of observations in that category. Moneyness is defined as $(K/S) - 1$, where K is the strike price and S is the spot price. Also shown is the average absolute percentage mispricing for the entire sample. Number of calls = 4578 (average bid–ask spread of 61.2 cents) and number of puts = 6890 (average bid–ask spread of 60.3 cents). $SV_{\rho \neq 0}$ and $SV_{\rho=0}$ are the SV models with non-zero and zero correlation respectively and BS is the Black–Scholes model.

maturity. Under the constrained SV model (i.e., $\rho = 0$), the average absolute percentage mispricing for the entire sample of calls is 7.16%, while under the unconstrained model ($\rho \neq 0$), the corresponding mispricing for calls is 5.29%. For put options, the improvement in mispricing under the unconstrained model is even better; the absolute percentage is 10.09% under the constrained model and 5.3% under the unconstrained model. Looking at Table 3, one finds that the pricing differences between the constrained and the unconstrained model are especially significant for out-of-the-money puts and out-of-the-money calls. For example the average absolute percentage mispricing for deep out-of-the-money calls ($K/S - 1 > 0.04$ and 0–3 months maturity) is

99% for the constrained SV model and 35% for the unconstrained SV model. Similarly, for deep out-of-the-money puts ($K/S - 1 < -0.04$ and 0–3 months maturity) the average absolute mispricing is 61% if correlation is set to zero and 18% if we allow for correlation. Also we find that the mispricing for deep out-of-the-money options decreases with maturity.

Interestingly enough, the BS model, with a daily estimation of an implied volatility outperforms the constrained SV model in terms pricing, out-of-sample, deep out-of-the-money calls and puts. This runs contrary to the typically held belief that a SV model (even one with zero correlation between returns and volatility) is always more desirable than the simple BS model. One may question whether any intraperiod variation in volatility which would work against the maintained assumption of a constant volatility between 2:00 and 2:20 p.m. would alter our results because the SV model is penalized by such intraperiod variation. However, the same estimation was done on two more samples in which intraperiod volatility may be higher, one created between 9:00 and 9:30 a.m. (towards the opening of the day) and the other created between 2:30 and 3:00 p.m. (towards the close of the day) and the results are virtually identical to the ones reported here. This suggests that our results are not sensitive to any intraperiod variation in volatility.

If one calculates the raw (not absolute) mispricings (market price – model price) for out-of-the-money options, one finds that both versions of the SV model underprice out-of-the-money puts and overprice out-of-the-money calls, although the amount of mispricing is much lower in the unconstrained model. The direction of mispricing for out-of-the-money options is consistent with an unusual high concentration of probability mass in the left tail of the risk-neutral distribution of the index returns that could have resulted from a crash-phobia regarding the market and have also been reported by Dumas et al. (1998) for S&P 500 index options. Although it is possible that incorporating a jump in the asset returns process (Bates, 1996a, b) would make the risk-neutral distribution more leptokurtic, the absence of a sufficient number of jumps in our sample would render the power from econometric estimation very low.

4.3. *Out-of-sample hedging*

The hedging results are reported in Table 4 from 13 April 1992 to 28 October 1992.¹⁶ Figs. 2 and 3 show the time series of the daily normalized changes in the hedge portfolios for out-of-the-money calls and puts respectively.

¹⁶ Rebalancing intervals of three days and a week were also used. However, as expected, the results deteriorated sharply with an increase in the rebalancing interval for both versions of the model.

Table 4

Shows the mean and root mean squared error (RMSE) of the daily normalized changes of the hedge portfolios from 13 April 1992 to 28 October 1992 for different classes of options under daily rebalancing

Option type	Mean		RMSE		V
	$SV_{\rho=0}$	$SV_{\rho \neq 0}$	$SV_{\rho=0}$	$SV_{\rho \neq 0}$	
Deep in-the-money calls	5.07×10^{-4}	4.99×10^{-4}	1.13×10^{-3}	1.07×10^{-3}	1.32
In-the-money calls	2.05×10^{-4}	2.17×10^{-4}	7.65×10^{-4}	6.92×10^{-4}	0.8
Near-the-money calls	7.7×10^{-5}	10.6×10^{-5}	3.24×10^{-4}	3.12×10^{-4}	1.23
Out-of-the-money calls	2.14×10^{-4}	1.25×10^{-4}	4.66×10^{-4}	3.45×10^{-5}	2.95
Deep out-of-the-money calls	-1.78×10^{-3}	0.59×10^{-3}	3.22×10^{-3}	0.82×10^{-3}	2.18
0-3 month calls	3.32×10^{-4}	3.34×10^{-4}	9.70×10^{-4}	8.95×10^{-5}	1.17
>3 month calls	4.32×10^{-4}	4.45×10^{-4}	9.83×10^{-4}	10.08×10^{-4}	1.56
Deep in-the-money puts	9.4×10^{-4}	7.96×10^{-4}	2.05×10^{-3}	2.08×10^{-3}	1.07
In-the-money puts	3.97×10^{-4}	2.32×10^{-4}	1.13×10^{-3}	1.18×10^{-3}	0.32
Near-the-money puts	8.7×10^{-5}	13.5×10^{-4}	10.25×10^{-4}	10.46×10^{-4}	1.15
Out-the-money puts	1.23×10^{-4}	1.08×10^{-5}	7.87×10^{-4}	2.05×10^{-4}	2.32
Deep out-of-the-money puts	0.3×10^{-3}	-2.7×10^{-3}	3.82×10^{-3}	0.52×10^{-3}	2.19
0-3 month puts	2.58×10^{-4}	2.03×10^{-4}	8.28×10^{-4}	7.99×10^{-4}	1.18
>3 month puts	5.7×10^{-4}	7.51×10^{-4}	9.16×10^{-4}	8.84×10^{-5}	1.17

Daily mean return of the S&P 500 index = 0.0002.

Standard deviation of the daily S&P 500 index return = 0.006.

An option is near-the-money if $|(S - K)/S| \leq 0.01$. A call is deep out-of-the money if $(K - S)/S > 0.04$, out of the-money if $0.01 < (K - S)/S \leq 0.04$. A call is deep in-the-money if $(S - K)/S > 0.04$ and in-the-money if $0.01 < (S - K)/S \leq 0.04$. Similarly the moneyness for puts is defined. V is the GMM test statistic for comparing the mean squared errors as described in the text and is distributed asymptotically as a standard normal. $SV_{\rho \neq 0}$ and $SV_{\rho=0}$ are the SV models with non-zero and zero correlation respectively.

We find that allowing for correlation between returns and volatility leads to significant (statistically) lower mean squared errors of the hedge portfolio only for out-of-the-money options (both calls and puts). This is consistent with the improvement in pricing as most of the pricing improvements after accounting for correlation were observed in out-of-the-money options. Since out-of-the-money puts are heavily traded in the index options market, this hedging result has some practical significance. Although not reported here, it is also true that the unconstrained SV model improves over the BS model for out-of-the-money options, but not for other option categories. However, there is no discernible difference between the constrained SV model and the BS model in terms of hedging for any of the option categories.

The hedging results suggest that failure to take into account the risk of a randomly changing volatility can result in poor hedges in the index options market for some very actively traded options. An important feature of the unconstrained SV model is that not only does this model allow for volatility

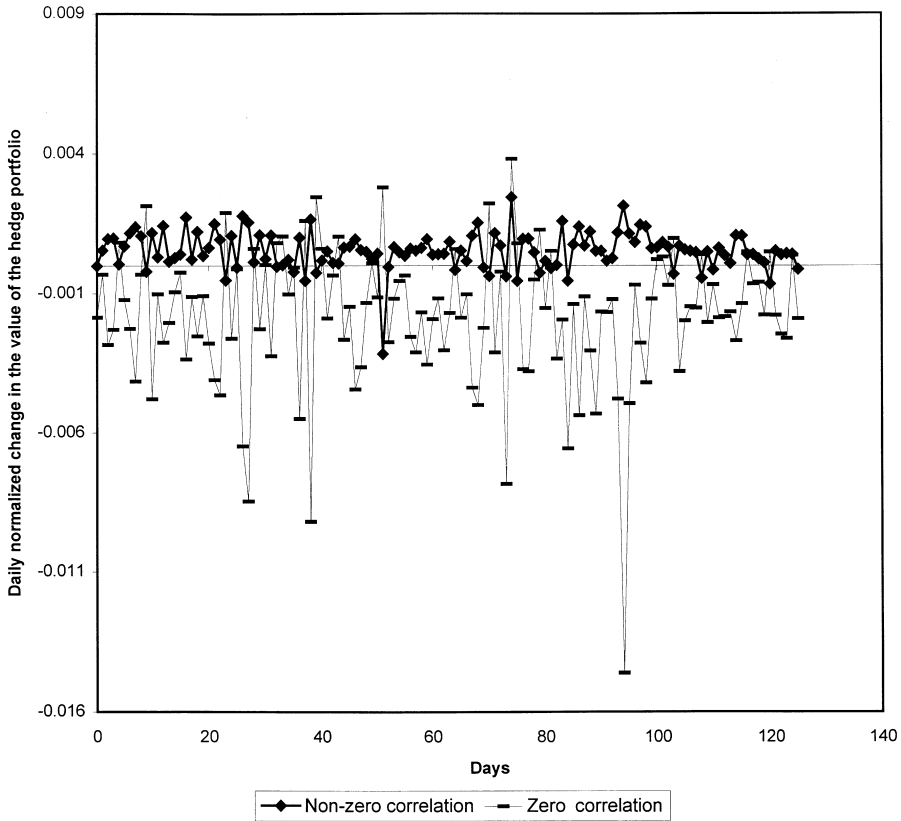


Fig. 2. Shows the daily normalized change in the value of the hedge portfolio for deep-out-of-the-money calls under both versions of the stochastic volatility model from 13 April 1992 to 28 October 1992.

shocks that are different from the shocks to the returns process, but volatility and spot index returns are negatively correlated (an often documented feature of equity indices) allowing this model to capture the changes in the index returns resulting from volatility shocks.

5. Conclusion

We looked at how the incorporation of a non-zero correlation between asset returns and volatility impacts pricing and hedging in a stochastic volatility environment. The stochastic volatility option pricing model developed by Heston (1993) that gives closed form solutions for option prices and hedge ratios was

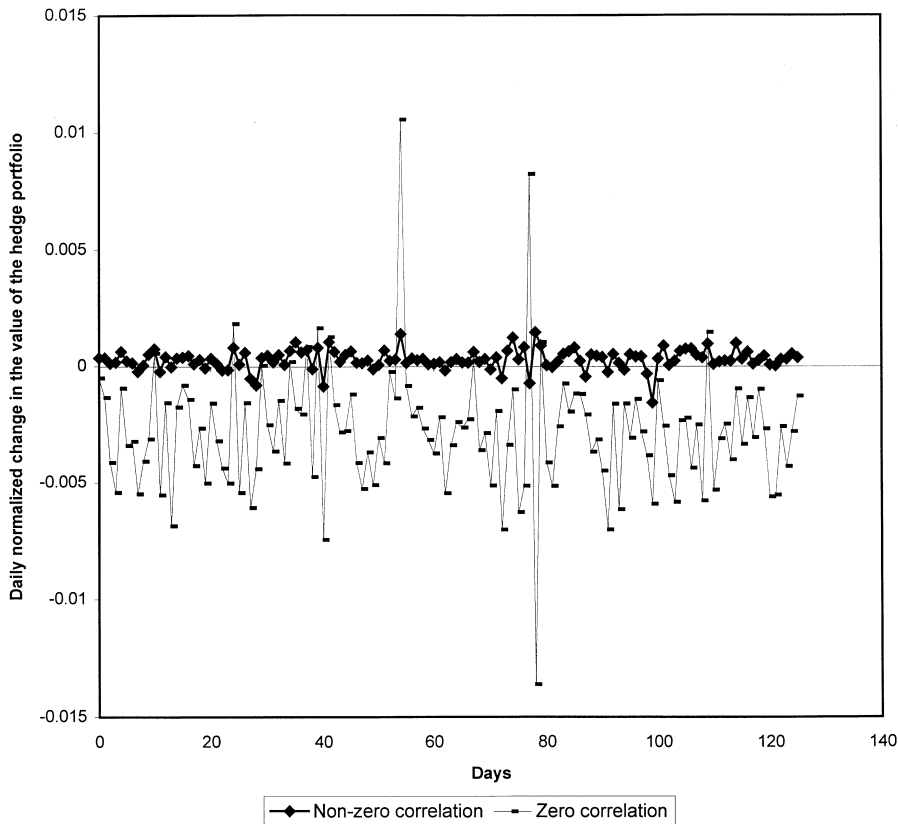


Fig. 3. Shows the daily normalized change in the value of the hedge portfolio for deep-out-of-the-money puts under both versions of the stochastic volatility model from 13 April 1992 to 28 October 1992.

used for this purpose. In order to avoid the potential inconsistency of using either Black–Scholes implied volatilities in a stochastic volatility model or estimating the time invariant parameters of the stochastic volatility model daily, the unobservable time varying volatility was estimated jointly with the other time invariant parameters, using both cross-sectional and time series data on options.

Substantial improvements in mispricing of out-of-the-money calls and puts are achieved by accounting for a non-zero correlation between returns and volatility in a stochastic volatility model in the index options market. Surprisingly, the widely used version of the Black–Scholes model in which an implied volatility is estimated daily outperforms the zero correlation version of the stochastic volatility model in terms of pricing (out-of-sample) deep-out-of-the-money options. However, the non-zero correlation version of the stochastic volatility

model outperforms the Black–Scholes model, both in terms of overall mispricing and pricing deep-out-of-the-money options. As far as hedging is concerned, after taking into account the transactions costs (bid–ask spreads) in the index options market and using S&P 500 futures to hedge, the stochastic volatility model results in lower mean squared errors for out-of-the-money options than the Black–Scholes model only if correlation is not constrained to be zero.

Future research can be directed towards developing stochastic volatility models that are easier to estimate and can also overcome the inconsistency between the parameters implied from option prices and the same set of parameters estimated from the time series of asset returns and volatility.

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Appendix A

The purpose of this appendix is to derive a formula for the half-life of the square root volatility process.

If spot variance of asset returns follows the process prescribed in Eq. (2) and $s - t \equiv \tau$ then (see Cox et al., 1985)

$$E[v_s|v_t] = v_t e^{-\kappa\tau} + \theta(1 - e^{-\kappa\tau}). \quad (\text{A.1})$$

If τ measures the half-life of the volatility process, it implies that half of the volatility shock that occurred at time t has been dissipated by time s . Since θ measures the long run variance, the expected variance at time s , conditional on the information at time t should equal $\theta - (\theta - v_t)/(2) = (\theta + v_t)/(2)$. Therefore,

$$\frac{v_t + \theta}{2} = v_t e^{-\kappa\tau} + \theta(1 - e^{-\kappa\tau}). \quad (\text{A.2})$$

Rearranging, we get,

$$v_t \left(\frac{1}{2} - e^{-\kappa\tau} \right) = \theta \left(\frac{1}{2} - e^{-\kappa\tau} \right). \quad (\text{A.3})$$

If the above equation is to hold for all values of v_t it is required that $1/2 - e^{-\kappa\tau} = 0$. This in turn yields $\tau = (\ln(2)/\kappa)$.

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